

## Question

A certain basic salt **A** of metal **M** is a bright green crystal. 2.524 g of this salt was placed in an inert atmosphere and heated to 148 °C, resulting in a weight loss of 12.19%, corresponding to the loss of all crystal water; continued heating to 233 °C, resulted in a weight loss of 9.26%; continued heating above 300 °C to constant weight, producing a gas with a density of 1.80 g/L (25 °C, 101.3 kPa). The composition of the remaining solid is an oxide of **M**, with a mass of 1.603 g. Let a be the column number of **M** in the periodic table, and b be the total number of ions in **A**, then a/b is approximately

**A.** All other options are incorrect

**B.** 0.41

**C.** 0.59

**D.** 0.69

**E.** 0.77

**F.** 0.91

**G.** 0.98

**H.** 1.23

**I.** 1.46

**J.** 1.65

**K.** 1.88

**L.** 2.01

**M.** 2.47

**N.** 1.06

**O.** 1.33

**P.** 0.33

## Answer

Correct Answer: **E**

## Detailed Explanation

From  $pM = \rho RT$ , it can be known that the molecular weight of the gas lost in the third step is 44.0, which is  $\text{CO}_2$

### CHECKPOINT

1 PTS

The gas lost in the third step is  $\text{CO}_2$

**A** is a basic carbonate, and the first two steps can only be water loss. Assume that the first step loses crystal water and the second step loses hydroxyl water.

Moles of crystal water  $n_1(\text{H}_2\text{O}) = 2.524 \text{ g} \times 12.19\% / 18.02 \text{ g} \cdot \text{mol}^{-1} = 0.01707 \text{ mol}$

### CHECKPOINT

0.5 PTS

Crystal water is 0.01707 mol

Moles of hydroxyl water  $n_2(\text{H}_2\text{O}) = 2.524 \text{ g} \times 9.26\% / 18.02 \text{ g} \cdot \text{mol}^{-1} = 0.01297 \text{ mol}$

## CHECKPOINT

0.5 PTS

Hydroxyl water is 0.01297 mol

$$n(\text{CO}_2) = [2.524 \text{ g} \times (1 - 12.19\% - 9.26\%) - 1.603 \text{ g}] / 44.01 \text{ g} \cdot \text{mol}^{-1} = 0.008625 \text{ mol}$$

## CHECKPOINT

0.5 PTS

0.008625 mol  $\text{CO}_2$  is released in the third step

Assuming that **M** in the remaining solid has a valence of +2, the number of moles of **MO** (corresponding to  $\text{CO}_2$  and hydroxyl water) is:  $n(\text{MO}) = 0.01297 \text{ mol} + 0.008625 \text{ mol} = 0.02160 \text{ mol}$

## CHECKPOINT

0.5 PTS

**MO** is 0.02160 mol

The molar ratio of crystal water, hydroxyl water,  $\text{CO}_2$ , and **M** is 4:3:2:5, so the chemical formula of **A** is  $\text{M}_5(\text{CO}_3)_2(\text{OH})_6 \cdot 4\text{H}_2\text{O}$

## CHECKPOINT

1 PTS

The chemical formula of **A** is  $\text{M}_5(\text{CO}_3)_2(\text{OH})_6 \cdot 4\text{H}_2\text{O}$

The molar mass of **MO** is  $1.603 \text{ g} / 0.02160 \text{ mol} = 74.21 \text{ g} \cdot \text{mol}^{-1}$ , **M** is Ni

## CHECKPOINT

0.5 PTS

**M** is Ni

If **M** has other valences, there are no reasonable results.

a=10, b=13, a/b=0.77, select E

## CHECKPOINT

0.5 PTS

a/b=0.77